

1. PROJECT OVERVIEW / INTRODUCTION

Customer support teams receive a large number of support tickets every day. Manually identifying the priority of each ticket and assigning it to the appropriate support agent can take time and may lead to delays in handling critical customer issues.

The **Customer Support Ticket Priority Prediction and Automated Assignment System using Agentforce** is designed to automate the analysis and prioritization of customer support tickets using Salesforce and Agent force. The system analyzes the ticket description and identifies whether the ticket should be classified as **High, Medium, or Low priority**.

The solution uses a Salesforce custom object called **Support Ticket Intelligence** to store and manage ticket information. A Salesforce **Auto-Launched Flow** processes the ticket information and performs priority analysis. The **Agentforce** agent uses this automated process to analyze customer support requests and provide the appropriate priority and assignment information.

When a ticket is identified as high priority, the system can initiate urgent-ticket handling and assign the ticket to an appropriate senior support resource. This helps reduce manual effort and supports faster handling of critical customer issues.

Project Objectives

- To automate customer support ticket priority classification.
- To identify high-priority customer issues quickly.
- To reduce manual ticket-prioritization effort.
- To support automatic assignment of tickets to appropriate support agents.
- To use Agentforce and Salesforce Flow for intelligent support-ticket automation.
- To improve the handling of urgent customer support requests.

Technologies Used

- Salesforce CRM
- Agentforce
- Salesforce Flow
- Salesforce Custom Objects
- Salesforce Tasks

The project user story and system approach are defined in the supplied project document.

2. IDEATION PHASE

2.1 Problem Statement

PS-1

I am (Customer): Support Agent

I'm trying to: Focus on critical tickets first.

But: Tickets are not always prioritized automatically.

Because: Ticket prioritization can be manual.

Which makes me feel: Frustrated.

PS-2

I am (Customer): Customer

I'm trying to: Get urgent issues resolved quickly.

But: Critical issues may wait behind lower-priority tickets.

Because: Support teams handle large ticket volumes.

Which makes me feel: Concerned.

PS-3

I am (Customer): Support Manager

I'm trying to: Assign tickets to the right support level.

But: Assignment may require manual decisions.

Because: Ticket urgency is not consistently analyzed.

Which makes me feel: Overwhelmed.

PS-4

I am (Customer): Support Agent

I'm trying to: Identify urgent issues from descriptions.

But: Important keywords can be missed.

Because: Ticket descriptions contain different urgency signals.

Which makes me feel: Uncertain.

PS-5

I am (Customer): Support Manager

I'm trying to: Reduce SLA-related risk.

But: Older unresolved tickets may remain unnoticed.

Because: SLA risk needs additional checking.

Which makes me feel: Concerned.

These five problem statements are based on the Problem Statement section of the supplied project template.

2.2 Empathy Map Canvas

User Persona: Support Agent

Think & Feel

The support agent wants critical issues to be identified quickly so that urgent customer problems receive proper attention.

Hear

The support agent receives customer issue descriptions and support requests.

See

The support agent sees ticket status, issue type, description, priority, assignment information, and SLA risk.

Say & Do

The support agent reviews tickets, handles assigned cases, and focuses on urgent customer issues.

Pain

- Manual ticket prioritization.
- Important urgency signals may be missed.
- Delays in handling critical issues.
- Difficulty in consistently identifying urgent tickets.

Gain

- Automatic priority classification.
- Clearer ticket assignment.
- Automatic urgent-task creation for high-priority issues.
- Reduced manual effort.
- Faster identification of critical tickets.

The Empathy Map Canvas and Support Agent persona follow the project template.

2.3 Brainstorming & Idea Prioritization

During the brainstorming stage, different approaches were considered to improve the process of handling customer support tickets.

Ideas Considered

1. Manual Ticket Registers

Support agents could manually record and prioritize customer support tickets. However, this approach requires continuous manual effort and may result in inconsistent prioritization.

2. Spreadsheet-Based Ticket Tracking

Support tickets could be maintained and tracked using spreadsheets. Although this provides basic organization, it does not provide intelligent ticket analysis or automated assignment.

3. Standalone Support Portal

A separate support portal could be developed for managing customer tickets. This could provide a dedicated interface but would require additional development and integration.

4. Salesforce CRM with Agentforce

Salesforce CRM combined with Agentforce can provide automated ticket analysis, priority classification, assignment support, and workflow automation within a single platform.

Selected Idea

The selected approach is **Salesforce CRM with Agentforce**.

The solution combines Salesforce data management, Salesforce Flow automation, and Agentforce-based analysis to support automatic priority classification and ticket assignment.

Key Reasons for Selecting the Idea

- Flow-based automation.
- AI-assisted Agentforce analysis.
- Automatic priority classification.
- Automatic handling of high-priority tickets.

- Reduced manual effort.
- Integration with Salesforce CRM.

The considered ideas and selected Salesforce CRM with Agentforce approach are specified in the project's brainstorming and idea-prioritization section.

3. REQUIREMENT ANALYSIS

3.1 Existing System

In the existing customer support process, support tickets are reviewed manually by support agents or managers. The priority of a ticket is generally determined by reading the ticket description and understanding the urgency of the issue.

Manual ticket handling can create difficulties when a large number of tickets are received. Important urgency indicators may be missed, and assigning tickets to the appropriate support level may require additional manual effort.

The major limitations of the existing system are:

- * Manual ticket prioritization.
- * Difficulty in identifying urgent issues quickly.
- * Possibility of missing important urgency signals.
- * Manual assignment of support tickets.
- * Increased effort for support agents.
- * Risk of delays in handling critical customer issues.

3.2 Proposed System

The proposed system uses **Salesforce, Agentforce, and Salesforce Flow** to support automated customer support ticket analysis and assignment.

The system stores ticket information in the **Support Ticket Intelligence** custom object. The ticket description is analyzed through a Salesforce Flow and Agentforce action. Based on the identified urgency, the ticket is classified into **High, Medium, or Low priority**.

High-priority tickets can trigger urgent-ticket handling and assignment to an appropriate senior support resource.

Proposed System Workflow

1. Customer support ticket information is available in Salesforce.
2. The system retrieves the relevant customer and ticket information.
3. The ticket description is analyzed.
4. The system identifies urgency indicators.
5. The ticket is classified as High, Medium, or Low priority.
6. The appropriate support assignment is determined.
7. High-priority tickets can trigger urgent-task handling.
8. The result is provided through the Agentforce interaction.

The project's user story describes this Salesforce + Agentforce workflow and priority classification approach.

3.3 Functional Requirements

The system should provide the following functional capabilities:

FR-1: Ticket Information Management

The system should store and manage customer support ticket information using the Salesforce **Support Ticket Intelligence** custom object.

FR-2: Ticket Description Analysis

The system should analyze the description of a customer support ticket to identify urgency-related information.

FR-3: Priority Classification

The system should classify support tickets into:

- High
- Medium
- Low

FR-4: Automated Ticket Assignment

The system should support assigning tickets to an appropriate support resource based on the identified priority.

FR-5: High-Priority Ticket Handling

When a ticket is identified as high priority, the system should support urgent-ticket handling, including creation of an urgent task where applicable.

FR-6: Agentforce Interaction

The Agentforce agent should be able to invoke the ticket-priority analysis action and provide the analysis result.

FR-7: Flow Automation

Salesforce Flow should process the ticket information and perform the required automation.

3.4 Non-Functional Requirements

Performance

The system should process ticket information efficiently and provide the priority-analysis result without unnecessary manual processing.

Usability

The system should provide a simple interaction through Agentforce so that support users can request ticket analysis without performing multiple manual steps.

Reliability

The automation should consistently apply the defined priority-classification logic to ticket descriptions.

Maintainability

The system should be easy to maintain and update using Salesforce configuration, Flow, and Agentforce components.

Security

The system should operate within Salesforce's configured user permissions and access controls.

3.5 Hardware Requirements

The project does not require specialized hardware.

Minimum requirements include:

- Computer or laptop.
- Stable Internet connection.
- Modern web browser.
- Keyboard and mouse.
- Sufficient system resources to access Salesforce and development tools.

3.6 Data Requirements

The system requires customer support ticket information for analysis.

Important ticket information includes:

- Ticket Number
- Customer
- Contact
- Issue Type
- Description
- Priority Level
- Status
- Assigned To
- SLA Breach Risk
- Created Date
- Resolution-related information

4. SYSTEM DESIGN

4.1 System Architecture

The proposed system is designed using Salesforce CRM, Salesforce Flow, and Agentforce. The architecture connects customer support ticket information with an automated priority-analysis process.

The **Support Ticket Intelligence** custom object is used to store and manage customer support ticket information. Salesforce Flow processes the ticket information and performs the required automation. Agentforce provides an intelligent interface through which the support ticket analysis action can be invoked.

The main components of the system are:

- Salesforce CRM
- Support Ticket Intelligence custom object
- Salesforce Auto-Launched Flow
- Agentforce Agent
- Support Ticket Priority Analysis action
- Automated task handling
- Support agent assignment

The overall architecture follows the project document's defined Salesforce, Flow, and Agentforce components.

4.2 System Workflow

The system follows a sequence of steps to analyze and handle customer support tickets.

Step 1 — Customer Support Ticket

A customer support ticket contains information such as the customer, issue type, and description of the problem.

Step 2 — Ticket Information Retrieval

The system retrieves the relevant customer and support ticket information from Salesforce.

Step 3 — Description Analysis

The ticket description is analyzed to identify urgency-related words and information.

Step 4 — Priority Classification

The system determines the appropriate priority level.

The possible priority levels are:

- High Priority
- Medium Priority
- Low Priority

Step 5 — Automated Assignment

Based on the analysis, the system determines the appropriate support assignment.

Step 6 — Urgent Ticket Handling

If the ticket is identified as high priority, the system supports urgent-ticket handling and can create an urgent task for appropriate support personnel.

Step 7 — Agentforce Response

Agentforce provides the analysis result to the user through the configured ticket-priority analysis action.

4.3 System Components

Support Ticket Intelligence

Support Ticket Intelligence is the Salesforce custom object used to store customer support ticket information and the results of ticket analysis. It provides a structured location for managing information required by the automation.

Salesforce Flow

An Auto-Launched Salesforce Flow is used to process the support ticket information.

The Flow performs activities such as retrieving the relevant account and ticket information, analyzing the ticket description, determining the priority, and preparing the required assignment and action information.

Agentforce Agent

The Agentforce agent provides the conversational interface for requesting support ticket analysis.

The agent can invoke the configured **Analyze Support Ticket Priority** action.

Support Ticket Priority Analysis Action

The Agentforce action connects the Agentforce agent with the Salesforce Flow.

The action receives the required ticket/customer information and returns the analysis results such as:

- Account information
- Ticket information
- Priority level
- Assigned support resource
- Action message

4.4 Priority Analysis Logic

The system analyzes the ticket description to identify urgency indicators.

High-priority examples include descriptions containing terms such as:

- Urgent
- Not working
- Failure

Medium-priority examples include terms such as:

- Issue
- Slow
- Delay

If no defined urgency indicators are identified, the ticket can be classified as Low priority.

This approach allows the system to provide a consistent initial priority classification based on the ticket description.

4.5 Data Flow

The data flows through the system in the following sequence:

Customer Support Ticket → Salesforce Support Ticket Intelligence → Salesforce Flow → Ticket Description Analysis → Priority Classification → Assignment / Urgent Handling → Agentforce Response

The ticket information is first stored in Salesforce. The Flow processes the relevant information and performs the priority analysis. The result is then returned to the Agentforce interaction.

4.6 System Output

After the ticket is analyzed, the system provides information such as:

- Ticket ID
- Priority Level
- Assigned Support Resource
- Action Message
- Urgent handling information when applicable

The output helps the support team understand the urgency of the ticket and take appropriate action.**5.**

PLANNING

5.1 Project Plan

The project is planned in different stages, starting from identifying the customer support problem and ending with testing and documentation.

5.2 Project Activities

- Problem identification and ideation
- Requirement analysis
- Salesforce object and field creation
- Salesforce Flow development
- Agentforce configuration
- Priority analysis and assignment automation
- Testing
- Documentation and GitHub submission

5.3 Project Milestones

Milestone 1: Ideation and requirement analysis

Milestone 2: Salesforce configuration and Flow development

Milestone 3: Agentforce integration and automation

Milestone 4: Testing, documentation, and final submission

5.4 Expected Outcome

The planned system should analyze customer support tickets, classify their priority, support appropriate assignment, and provide automated handling for high-priority tickets using Salesforce and Agentforce.

6. DEVELOPMENT / IMPLEMENTATION

6.1 Salesforce Setup

The project was developed using Salesforce CRM. A custom object named **Support Ticket Intelligence** was created to manage customer support ticket information.

6.2 Custom Object and Fields

The Support Ticket Intelligence object stores important ticket details such as customer, issue type, description, priority, status, and assignment information.

6.3 Salesforce Flow

An **Auto-Launched Flow** was developed to process support ticket information. The Flow retrieves the required ticket details, analyzes the description, determines the priority, and prepares the assignment and action results.

6.4 Agentforce Configuration

An Agentforce agent was configured with the **Analyze Support Ticket Priority action**. This action connects Agentforce with the Salesforce Flow to perform ticket analysis.

6.5 Priority Prediction

The system analyzes the ticket description and classifies the ticket as:

- **High Priority**
- **Medium Priority**
- **Low Priority**

Urgency-related keywords are used as indicators for the priority classification.

6.6 Automated Assignment

Based on the identified priority, the system provides the appropriate assignment information. High-priority tickets can trigger urgent-ticket handling and assignment to a senior support resource.

6.7 Implementation Result

The completed implementation connects Salesforce ticket data, Flow automation, and Agentforce to support automated ticket priority analysis and assignment.

7. TESTING

The system was tested using different customer support ticket descriptions to verify priority classification and automated handling.

The testing focused on:

- High-priority ticket identification.
- Medium and Low priority classification.
- Agentforce action execution.
- Assignment and urgent-task handling

8. RESULTS / SCREENSHOTS

The implemented system successfully connects Salesforce, Salesforce Flow, and Agentforce for customer support ticket analysis.

Screenshots of the Salesforce configuration, Flow, Agentforce setup, and final output are included to demonstrate the project implementation.

9. CONCLUSION

The project provides an automated approach for analyzing and prioritizing customer support tickets using Salesforce and Agentforce. It helps reduce manual prioritization and supports faster handling and assignment of critical tickets.

10. FUTURE ENHANCEMENT

The system can be enhanced by:

- Using advanced AI models for more accurate priority prediction.
- Adding multilingual ticket analysis.
- Improving SLA-risk prediction.
- Adding dashboards for support managers.

11. REFERENCES

- Salesforce Documentation
- Salesforce Trailhead
- Agentforce Documentation
- Project template and learning resources provided by the institution